7. FUNCTIONAL & PERFORMANCE TESTING

The application was tested end to end inside the Salesforce Developer Org. For each test, screenshots were captured to record the outcome. The tests covered the main functions of the system: creating records, enforcing validation rules, recording delivery updates, running flow automation, generating invoices, and displaying dashboards. Every scenario behaved as expected, confirming that the system works correctly and reliably.

Test Scenarios and Results

Test Area	Scenario Tested	Expected Result	Outcome
Record creation	Create customer, shipment, delivery agent, branch, delivery update, and invoice records	Records are saved and linked through their relationships	Verified
Validation rules	Save records with a blank email, an invalid phone number or pincode, a zero weight or price, or a past expected date	The save is blocked and the configured error message is shown	Verified
Delivery status update	Create a delivery update for a shipment	The related shipment's status changes to match the update	Verified
Flow automation	Trigger the Auto_Update_Shipment_Status flow	The flow runs automatically with no manual action	Verified
Invoice generation	Create an invoice for a delivered shipment	The invoice is saved with the correct amount, date, and payment status	Verified
Security	Sign in as users with different profiles and roles	Each user sees only the tabs, objects, and records allowed for their profile and role	Verified
Reports and dashboards	Open the Shipment Status Summary, Agent Performance, and Revenue by Branch reports and the dashboard	Charts display the current data correctly	Verified

Performance Observations

- Shipment status changes from delivery updates appeared on the shipment record straight away
- Reports and dashboards loaded quickly and refreshed with the latest data
- Navigation between the app's tabs and record pages was smooth

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Single source of truth:** customer, shipment, agent, branch, and invoice data live in one platform instead of scattered registers and spreadsheets
- **Less manual work:** flows and validation rules handle status updates and data checks automatically, cutting down on human error

- **Live visibility:** managers and customers can see where a shipment stands without chasing updates
- **Informed decisions:** reports and dashboards show delivery performance, shipment status, and revenue at a glance
- **Controlled access:** profiles and roles ensure each user sees only the data their job requires
- **Room to grow:** the cloud-based design supports additional branches, users, and records over time

Disadvantages

- **Platform lock-in:** the solution relies on Salesforce, so the organization depends on its licensing, limits, and release cycle
- **Setup effort:** objects, fields, flows, profiles, and roles must be configured carefully before the system is ready to use
- **Ongoing cost and training:** licences are a recurring expense, and staff need training to use the system well
- **Connectivity needed:** users require a reliable internet connection to work with the application

9. Conclusion:

The Courier Management System shows how a courier business can move from paper registers and disconnected spreadsheets to a single, automated platform built on Salesforce. Customers, shipments, delivery agents, branches, status updates, and invoices are all managed in one place, and validation rules and flows keep the data accurate while reducing repetitive work. Role-based security protects sensitive information, and the reports and dashboards give managers a clear, current view of operations. Together these capabilities speed up deliveries, improve accuracy, and leave customers better informed and more satisfied.

10. FUTURE SCOPE

The system can be extended in several directions to make it even more useful:

- **Mobile tracking app:** a companion app for customers and delivery agents to book, track, and update shipments on the go
- **GPS integration:** live location of vehicles and parcels, displayed on a map
- **SMS and email notifications:** automatic alerts to customers at each stage of delivery
- **AI-based delivery optimization:** smarter route planning and agent assignment based on workload and delivery history
- **Online payments:** payment gateway integration so invoices can be settled digitally

11. APPENDIX

- **Source Code:** the solution is built entirely through Salesforce configuration (objects, fields, validation rules, flows, profiles, and roles), so no custom code is involved
- **Dataset:** the data is stored in Salesforce custom objects: Customer, Shipment, Delivery Agent, Branch, Delivery Update, and Invoice
- **GitHub / Demo:** the working application is available in the Salesforce Developer Org used for this project

Thank You